



Building National Resilience Through Defense Entrepreneurship

Gary Ang

Professor of Practice (Defence Innovation and Applications)
NUS Enterprise

*"In today's world, it is more than just about equipping and transforming the SAF, it is also thinking hard about technology and thinking hard about defense supply chains and how we can be more resilient as a country... **it is important for the country to build up its own capabilities.**"*

PM Lawrence Wong, Mandai Hill Camp, April 2026

The Strategic Thesis

Three structural shifts have re-priced defense.

01

Warfare Dynamic

Unmanned offensives are cheap, attritable, and used at scale. The cost asymmetry has fundamentally changed the calculus of conflict.

02

Supply

Global markets may not supply what Singapore needs, when it needs it, at scale. Supply chain resilience is now a sovereign imperative.

03

Demand

Sovereign design + manufacturing now commands a premium over legacy primes. Nations are paying for control, not just capability.

- ① MINDEF (buyer/operator) and ST Engineering (prime integrator) remain load-bearing. What is missing is a **third feeder**: universities producing dual-fluent founders who build at venture speed and feed the national stack. **DVL closes that gap** — supplying MINDEF and the primes, not replacing them.

The Age of Asymmetry,

The disparity that connected defense to startups.

01

Cost-Exchange Inverted

Operation Spiderweb used cheap drones hidden in trucks to destroy billions in Russian strategic bombers on the tarmac.



02

Reach Without Mass

Small teams with autonomy, cyber tools and **off-the-shelf hardware** can hold complex infrastructure at risk from far away.

and an ever-changing **arms-control regime,**

Eroding guardrails

Today's export controls cannot keep up with advanced weapons: newer technology will proliferate faster, to more actors, slipping by the MTCR, NSG, Wassenaar rule-of-law orders, that are weakening.

Arriving in our neighbourhood

On 7 July 2026 India agreed to sell BrahMos cruise and Astra air-to-air missiles to Indonesia (~\$630M), the third Indo-Pacific BrahMos buyer after the Philippines and Vietnam.

Sovereign innovation that delivers **good, cheap and many**

Real example · Ukrainian Interceptors

Traditional air defense planners were in for a shock when the everyday FPV drone turned out to be the best weapon in the hangar - pivoting from hobby toy to a distributed, on-demand shield against loitering munitions. This grassroots shift bypasses rigid military bureaucracy, empowering a hyper-local network of workshops to out-innovate state-level defense contractors into mid-2026.

Good

Deploy nimble, hunter-killer quads directly at the front line - rendering multi-million dollar missile radars obsolete.

Cheap

Off-the-shelf plastic parts and cheap optical sensors completely replace the crushing cost of conventional anti-air missiles.

Many

Send open-source CAD blueprints straight to thousands of standard 3D printers.



The COVID Moment of Geopolitics

Markets work, until everyone
needs the same thing at once.

\$2.7T

Global military spending in 2024 - up
~9%, the sharpest annual rise in more
than three decades.

01

Access Is Not Supply

In crises, states hoard supplies, restrict
exports and override contracts.

02

Priority Follows Sovereignty

In crisis, suppliers hoard and may not
honour prior contracts.

03

Single Points of Failure

Any one shipping lane, export license or
component can gate a whole capability.



155mm shells in the Ukraine War

Ukraine is firing shells faster than can be supplied.

Can Europe catch up? - **CNN Sep 23**

Even today:

Ukraine over-subscription of 155mm shells in 2023 is
still being back-filled. **As of last week, the U.S. is still
struggling to keep up with their production goals.**

This chain was not resilient.

(**target** - 100,000/month) (**achieved** - 36,000/month)

Supply chains are strategic terrain

Defense tech - and the supply chains that produce it - now decide the balance of power, before a shot is fired.

01 Technology sets the balance

The decisive military edge is now generated in fast-moving, dual-use tech. Whoever fields it first shapes the balance of power.

02 It runs on concentrated chains

That edge depends on chips, minerals, components and capacity - each held by a handful of suppliers, often in one place.

03 Dependence becomes leverage

Control one input and you can throttle a rival without firing a shot. Your own dependency is the opening they exploit.

~90%

Semiconductors

The world's most advanced chips - made on one island (Taiwan).

~90%

Critical minerals

China's share of rare-earth refining - ~99% for the heavy ones.

F-35

Fifth-tier supplier

Deliveries paused in 2022 after Chinese-origin alloy was found in a magnet.

Why Resilience

A stockpile depletes. Resilience regenerates - and in a long fight, the flow beats the stock.

A Sustainable Flow



Regenerate

Replace losses at the tempo of the fight.

Adapt

Turn a new problem into a fielded answer in weeks.

Sustain

Repair and keep degraded systems in the fight.

Stay sovereign

Enough built at home that no one can switch you off.

A Depleting Stock



Depletes faster than any plan assumes.

You can't surge what you can't make.

One lane, licence or component gates it all.

A stockpile depreciates. An ecosystem compounds. Defense entrepreneurship is how a small state buys regenerative depth.

The Next Wave Is Already Here

Startups now build at the frontier the primes once owned alone.

01

SpaceX — f. 2002 · \$1T+

NASA + USAF/Space Force; Starlink \$900M DoD contract → commercial launch + Starlink broadband + maritime/aviation connectivity. **Moat:** reusable rocket physics took 10 years and billions to solve.

02

Anduril — f. 2017 · \$61B

Border surveillance, counter-drone, USSOCOM → Lattice OS as dual-use autonomy platform. **Moat:** Lattice is an operating system for autonomous systems — compounds with every customer.

03

Palantir — f. 2003 · \$341B

CIA seed funding → Gotham (defense) → Foundry (enterprise) → AIP (AI platform). **Moat:** 10 years of classified deployment hardened against adversarial conditions no commercial environment ever tests.

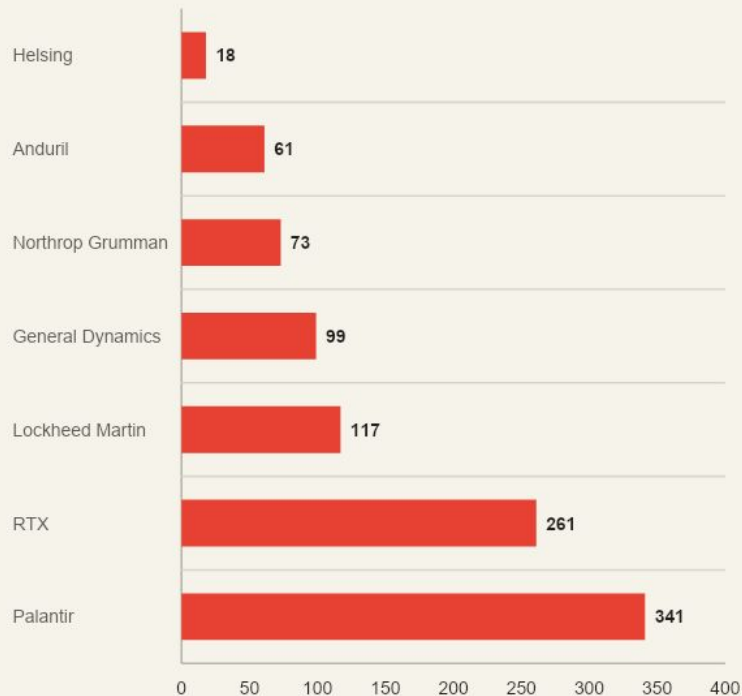
04

Helsing — f. 2021 · \$18B

Europe's defense-AI champion - proof the model is not just American.

Public market caps: 21 Jul 2026. Private valuations: Anduril Series H; Helsing Series E.

Selected valuations (\$bn), Jul 2026



Defense Is Now Investable

Four structural shifts changed the venture equation.

Durable Demand

States are raising budgets and rebuilding inventories. The buyer is no longer occasional.

Cheaper to Build

Compute, sensors, components and software can be sourced from broad commercial markets.

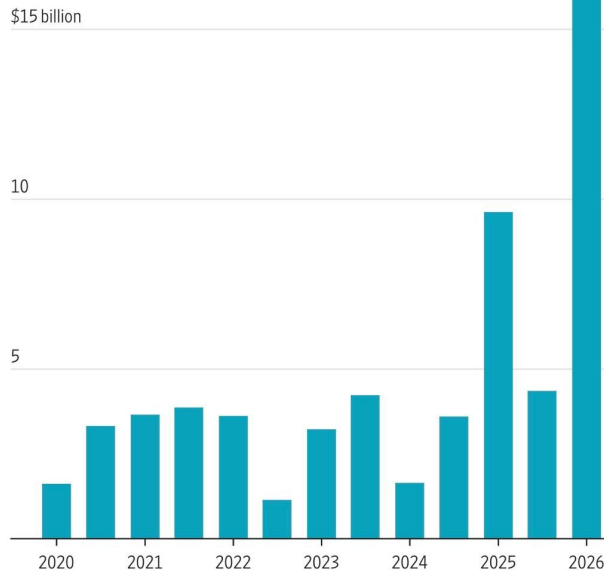
Faster Cycles

Software-defined products and modular hardware improve in months, not decades.

New Buying Paths

Pilots, challenges and rapid-capability offices open doors traditional programmes kept shut.

U.S. venture-capital investments in defense and aerospace startups, by half-year



Source: PitchBook

Request for Startups,
Summer 2026



Counter-Swarm Defense:

Developing a software-defined defense stack to neutralize cheap, autonomous drone swarms, moving away from expensive missile systems toward distributed, high-capacity interceptors.

Market Context: Defense at valuation of **\$750B** sits competitively alongside Cloud Services (\$616B), Telecom Equipment (\$658B), and Electric Vehicles (\$714B) — firmly in the tier of the world's most significant technology markets.

Singapore's Defense Innovation Stack, today.

Strong institutions. One missing feeder.

01

Mission Owners

MINDEF and the SAF set requirements, operate the systems and validate what actually works.

02

Depth and Scale

D50 and DSTA provide deep research and systems engineering. ST Engineering integrates, certifies and manufactures.

03

The Missing Feeder

A repeatable way to turn technical and operational talent into companies that supply the stack.

03

Credible Institutions fostering research talent

NUS, NTU and A*STAR turn out world-class engineers and scientists every year. The raw talent is already here.

Inaugural Defense Tech Hackathon Phase 1

Partners as
Strategic Anchors

EDTH | European
Defense
Tech Hub

NUS
National University
of Singapore

Enterprise

W | TUM

VENTURE
LABS

SINGAPORE DEFENSE TECH HACKATHON

SINGAPORE, SEPTEMBER 25-27



The Judges

On Sep 27, prizes will be presented by MG Kelvin Fan, Chief of Air Force (GOH).

Panel A

Panel A



MG Kelvin Fan
Chief of Air Force



Prof Quek Tong Boon
Former Chief Defence Scientist



Mr Timo Levo
Global Defence Lead, Accenture



Mr Daryl Lee
Head of Engineering, DSTA



Mr Mui Whye Kee
Group Technology Officer, DSTA

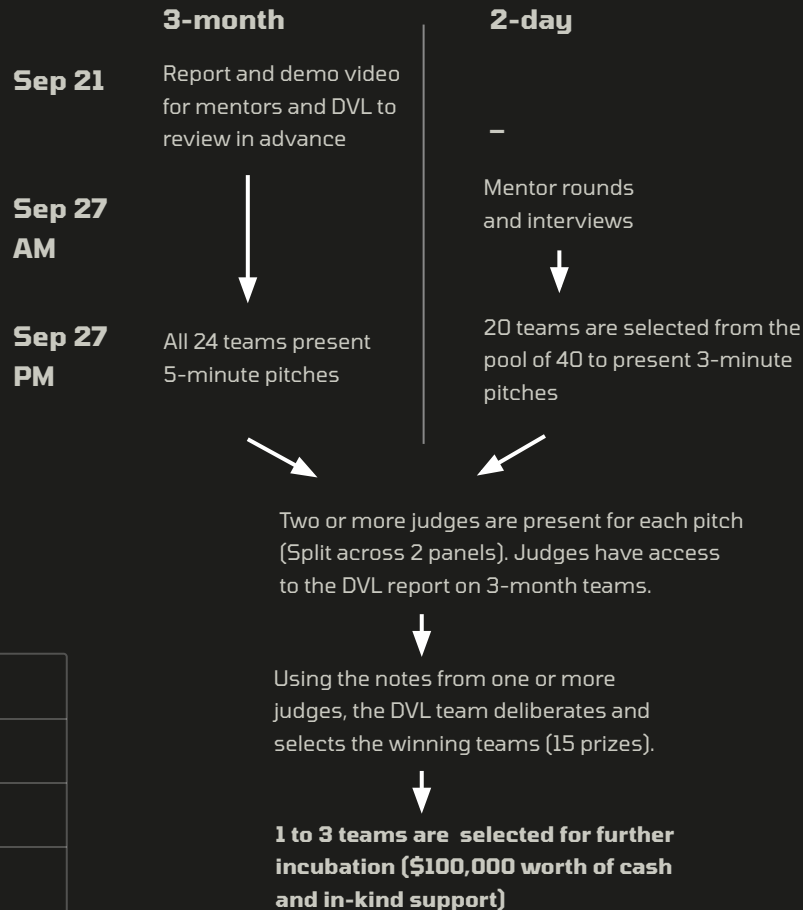


Mr Tan Peng Yam
Chief Defence Scientist

Judging Criteria

Criterion	Weight
Does it work? - shown through software/hardware demos	40%
Does the mission need it? - operational fit	30%
Can it be incubated? - team, economics, route to buyer	30%

Judging Flow



Prizes

		1st	2nd	3rd
3-month Teams	Main Prize	\$10,000	\$5,000	\$3,000
	The Lord of The Rings Prize	\$5,000	\$3,000	\$2,000
	2-day Support Prize	\$2,000	\$1,000	\$500
2-day Teams	Main Prize	\$8,000	\$4,000	\$2,000
	The Lord of The Rings Prize	\$3,000	\$2,000	\$1,000

The Lord of the Rings Prize (3-month) is split among 2 or more 3-month teams that collaborate to demonstrate a bigger play. Part of DVL’s mission is in letting startups help one another, like two parts of the same prime.

2-day Support Prizes are mapped to 3-month teams who support 2-day teams that win The Lord of The Rings Prize. Each 3-month team can back one 2-day team.

The 2-day Main Prize and 2-day Lord of The Rings Prize are mutually exclusive.

Leveraging the Singapore Defense Tech Hackathon in Phase 1 as a flagship event to bring together the world's most talented hackers and spot elite talent



September 25–27, 2026

48 hours of intensive innovation and rapid prototyping. Additional 3 month incubation track.



Strategic Location

National University of Singapore (NUS)

Asia-Pacific's premier Innovation hub



Global Reach

Leveraging EDTH's international network

Proven track record across 20+ events

1

Elite Talent Pipeline

Access 150+vetted builders—senior engineers, founders, and operators capable of executing under pressure. Participants average late 20s to mid-30s with significant industry experience.

2

Real-World Prototypes

Move beyond concepts to tangible outcomes. Teams deliver deployable prototypes within 48 hours, solving critical challenges in defense and security technology.

3

Strategic Partnerships

Define problem sets focused on Asia-Pacific security challenges including counter-UAS, maritime security, electronic warfare, and space defense resilience.

4

Proven ROI

Alumni have joined NATO DIANA 2026, raised millions in funding, and secured positions at leading defense tech companies. Direct access to hiring, deal flow, and collaboration opportunities.

The Pipeline

SDTH → DVL → INTERNATIONAL · STRUCTURED DEAL FLOW PIPELINE.

1

Stage 01 — SDTH

3-month track + 2-day hackathon

- Founder-matching across military operator / technical / commercial archetypes
- Industry-expert problem statements with real procurement intent
- Streaming of vetted talent into 3-month track

2

Stage 02 — DVL Incubation

6+ months

- Technical + venture mentorship
- SG + European defense testbed access
- ITAR / EAR / SGCA navigation
- Dual-geography prototyping (Singapore + Munich)
- Flexible curriculum tailored towards founder needs
- Milestone: seed-funding towards end of program

3

Stage 03 — International Validation

Ongoing

- Allied customers (US DoD, EU MoDs, dual-use)
- Production-scale partnerships
- Global VCs · Y Combinator · Thiel Fellowship
- Series A and beyond

Inaugural Cohort — Q2 2026

1000+

Signups

~90%

Domestic Share

170

2-day Track Participants

92

3-Month Track Participants

24

Long-Form (3-month) Projects launched

34

2-day Track Teams

Building the NUS Defense Tech Venture Lab under NUS Enterprise

This incubator acts as a "Dual-Fluency Bridge," transforming deep STEM research into global defense champions, addressing critical gaps, stemming from the changed nature of warfare because of AI, asymmetry in costs and capabilities, and space as new war fighting domain.

Strategic Context: The "Why"

New Reality: Geopolitics now functions as the "New COVID," mandating strategic self-reliance. This incubator addresses the **Innovation Gap**, enabling NUS spin-offs to join global leaders from Harvard, Stanford, MIT, Columbia and Cornell in acquiring elite talent and funding.

The Talent Magnet

The "Outlier" Draw: Top students are increasingly pivoting towards **"Tough Tech"** focused on crisis prevention and national resilience.



Dual-Fluency: A workforce skilled in high-level STEM and complex regulatory landscapes (e.g., ITAR/EAR compliance).

NUS Ecosystem: The "How"

A pillar of the Singapore Defense Technology Community, the program has a clear pipeline:

1. Defense Tech Hackathons (Knowledge Translation)
2. Defense Tech Incubator (Venture Creation)
3. Degree in Defense Tech (Academic Leadership)

Synergy with MINDEF

Evolution of Talent: From traditional feeders for DSTA, DSO and STE, to creating home-grown startups backed by world-class VCs that become Global champions.



The strategies the feeder must employ

For an effort to engineer defense entrepreneurs:

01 To be the Bridge Builder

People

Scientists, engineers, operators and founders fluent in both mission and technology.

Test and Production

Ranges, labs, qualified machines, tooling and local suppliers ready to make and prove things.

Knowledge and IP

Designs, models, software and materials recipes held locally, not just finished imports.

Procurement Paths

Clear routes from prototype to trial to limited deployment to scaled acquisition.

02 Fostering a System of Startups



Hackathon & Incubator Design

Guiding initial ventures towards a few high-potential themes provides a valuable north star. Our inaugural cohort focused on The Little Red Dome for air defence and The Neo-Battalion for evolving ground technology.



The Lord of the Rings Strategy

DVL thinks like a defense prime, getting startups with their singular tech to work together to create a unified capability that MINDEF and end-users want.

The strategies the feeder must employ

For an effort to engineer defense entrepreneurs:

03 The Joseph Schooling Method



Train with the best, abroad.

The world-class STEM talent in Singapore must be developed and then sent for further training abroad. Like Joseph Schooling, our defense talent must iterate and compete with the best in the world first - measure and improve against the highest bars - YC, a16z, the Thiel Fellowship, etc.

04 Implement a K-Pop Pipeline



Develop a pipeline to replicate success

Build a domestic system rigorous enough to consistently produce the Joseph Schoolings of defense. Like K-Pop, the edge is talent density and early scouting - spotting high-potential people young and concentrating them early via the hackathon and incubator pipeline.

The DVL Core Team

Strategic Leadership: Former heads of civil service, defense secretaries, and Temasek investors & operating partners – the institutional credibility that opens doors in Singapore's defense ecosystem.

- Government & policy access
- MINDEF / SAF relationships
- Investment network depth

1



BG (Ret) Gary Ang
Professor of Practice,
NUS

Ex-MD / Operating Partner
Temasek; Deputy
Secretary (Policy) MINDEF;
RSAF BG; ex-Cdr Tengah
Air Base.

2



Ben Cistecky
Venture Partner

Ex-Managing Partner
Certitude Holdings
(Temasek); 15+ years dual-
use, AI, fintech investing.

3



Noam Weisberg
Ops Tech Advisor

Ex-Senior Programme
Director, Israeli MoD; 20+
years mission-critical
aerospace / defense.

4



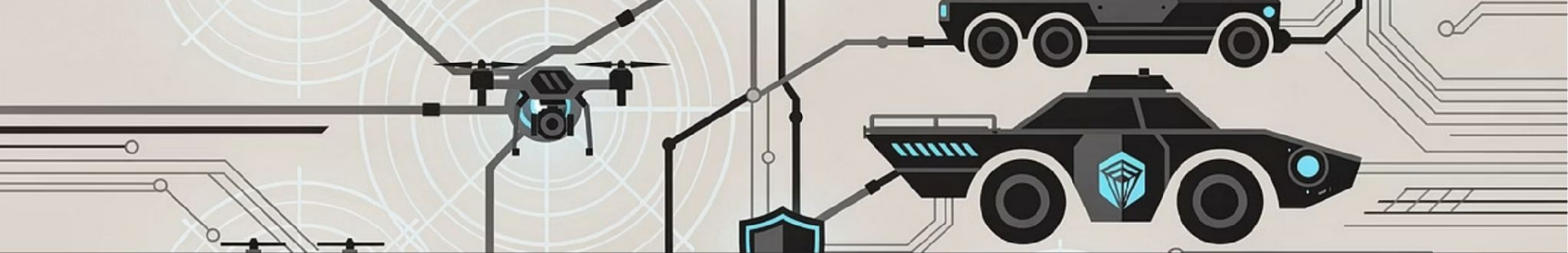
Stefan Taing
R&D Translation Advisor

CEO & Co-founder, Munich
Innovation Labs (acq.
Rohde & Schwarz);
industrial-engineering
PhD.



Peter Ho
Strategic
Mentor

Senior Advisor
Centre for
Strategic
Futures;
former Head
Civil Service;
Permanent
Secretary
Defence,
Foreign Affairs,
National
Security &
Intelligence
Coordination



Why this is about you:

Q&A

Commanders are at the ecosystem's core.

“A”

The Demand Signal

Surface what slows build-up, maintenance, training and operations.



The First Testers

Give startups the time, context and access needed for operational feedback.

U

The Absorbers

Recognise a fit-for-purpose capability and pull it into use.



The Invitation

Bring one real operational problem to the Singapore Defense Tech Hackathon and help test what emerges.

Gary Ang, Professor of Practice for Defence Innovation and Applications at NUS, Email: garyang@nus.edu.sg

Thank you for your time and consideration.